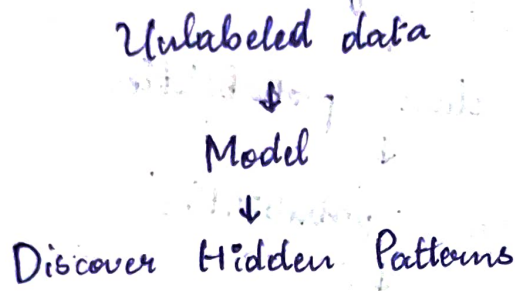


10/09/2026

UNSUPERVISED LEARNING + ENSEMBLE LEARNING + SEMI-SUPERVISED LEARNING + REINFORCEMENT LEARNING

1. Unsupervised learning - A machine learning approach where models learn patterns, structures or relationships from unlabeled data.

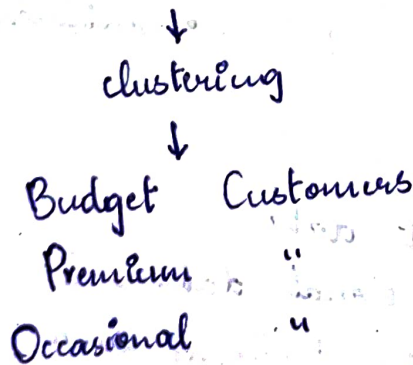


Main tasks :

- Clustering
- Dimensionality Reduction

2. Clustering - The process of grouping similar data points together without predefined labels.

example: Customer data



Common clustering models:

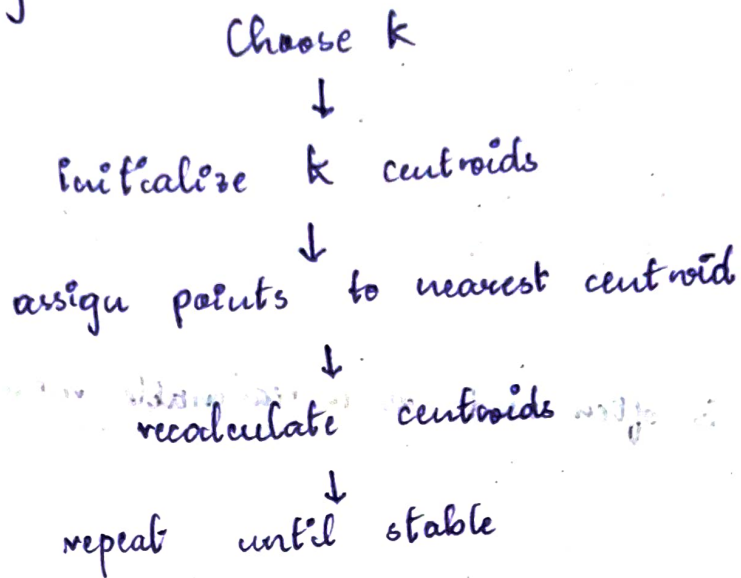
K-Means

Hierarchical Clustering

DBSCAN

8. K-Means Clustering - An unsupervised clustering algo that divides data into k groups by assigning each point to the nearest cluster centroid.

Working -



centroid centroid

K - number of clusters

Centroid - mean position of points in a cluster

Cluster - group of similar observations.

Inertia / WCSS - sum of squared distances between points and their cluster centroid.

$$WCSS = \sum_{k=1}^K \sum_{x_i \in S_k} \|x_i - \mu_k\|^2$$

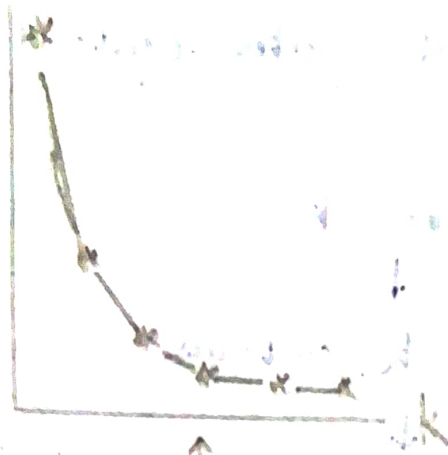
total no of clusters $\leftarrow$ K $\leftarrow$ $\sum_{k=1}^K$ $\leftarrow$ $\sum_{x_i \in S_k}$ $\leftarrow$ $\|x_i - \mu_k\|^2$ $\leftarrow$ individual point in cluster S_k $\leftarrow$ centroid of cluster k

$$\text{Euclidean } d \Rightarrow \|x_i - \mu_k\|^2 = \sum_{j=1}^D (x_{ij} - \mu_{kj})^2$$

x_{ij} = value of j^{th} feature for data point i

μ_{kj} = " " " for centroid k .

Choosing k - elbow method (extra information) -



The elbow point is often used as a reasonable value of k .

Advantages -

- Simple
- Fast
- Easy to understand
- Works well for compact spherical clusters.

Limitations -

- Sensitive to scaling
- " outliers
- Must choose k beforehand
- Struggles with irregularly shaped clusters.
- Sensitive to centroid initialization.

4. Hierarchical clustering - A clustering technique that creates a hierarchy of nested clusters.

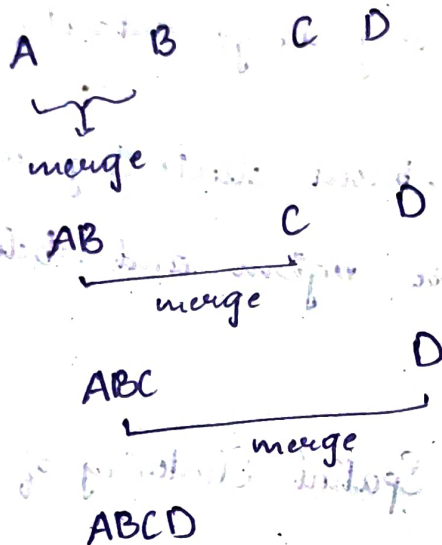
Main types:

- Agglomerative

- Divisive

→ Agglomerative -

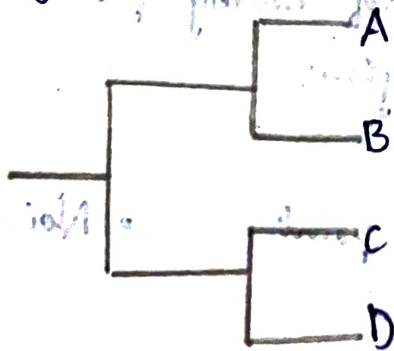
Starts with individual points and repeatedly merges them.



→ Divisive -

Starts with one large cluster and repeatedly splits it.

Dendrogram - A tree-like diagram showing how clusters are merged or separated.



Linkage Methods:

- Single linkage
- Complete "
- Average "
- Ward "

Advantages -

- No initial k required.
- Dendrogram provides visual interpretation
- Useful for smaller datasets

Limitations -

- Computationally expensive
- Sensitive to noise / outliers
- Difficult to scale to very large datasets.

5. DBSCAN - A density-based clustering algorithm that groups points in dense regions and labels isolated points as noise.

DBSCAN - Density-Based Spatial Clustering of Applications with Noise.

Main Parameters -

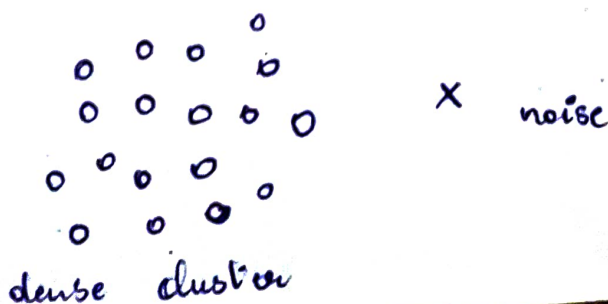
→ Epsilon ϵ - maximum distance for 2 points to be considered neighbors.

→ MinPts - minimum no. of nearby points required to form a dense region.

Point Types :

- Core point
- Border point
- Noise point

Representation -



Advantages -

- Does not require k
- Detects irregularly shaped clusters
- Can identify noise/outliers

Limitations -

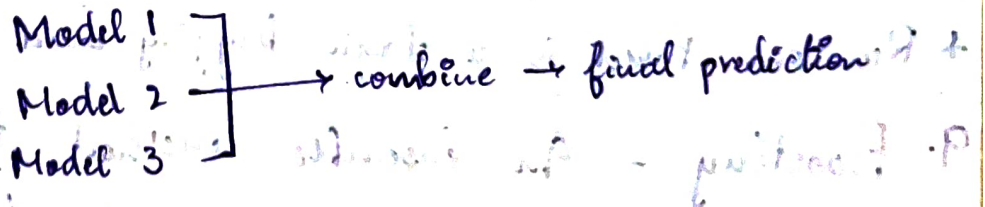
- Sensitive to ϵ and MinPts.
- Struggles when cluster densities vary
- Distance becomes less useful in high-dimensional data.

K-Means Vs Hierarchical Vs DBSCAN

Model	Main idea	Need k ?	Handles noise?
K-Means	Centroid-based	Yes	No
Hierarchical	Cluster hierarchy	Not initially	Limited
DBSCAN	Density-based	No	Yes

6. PCA - Recap

7. Ensemble learning - A technique that combines predictions from multiple models to produce a stronger final model



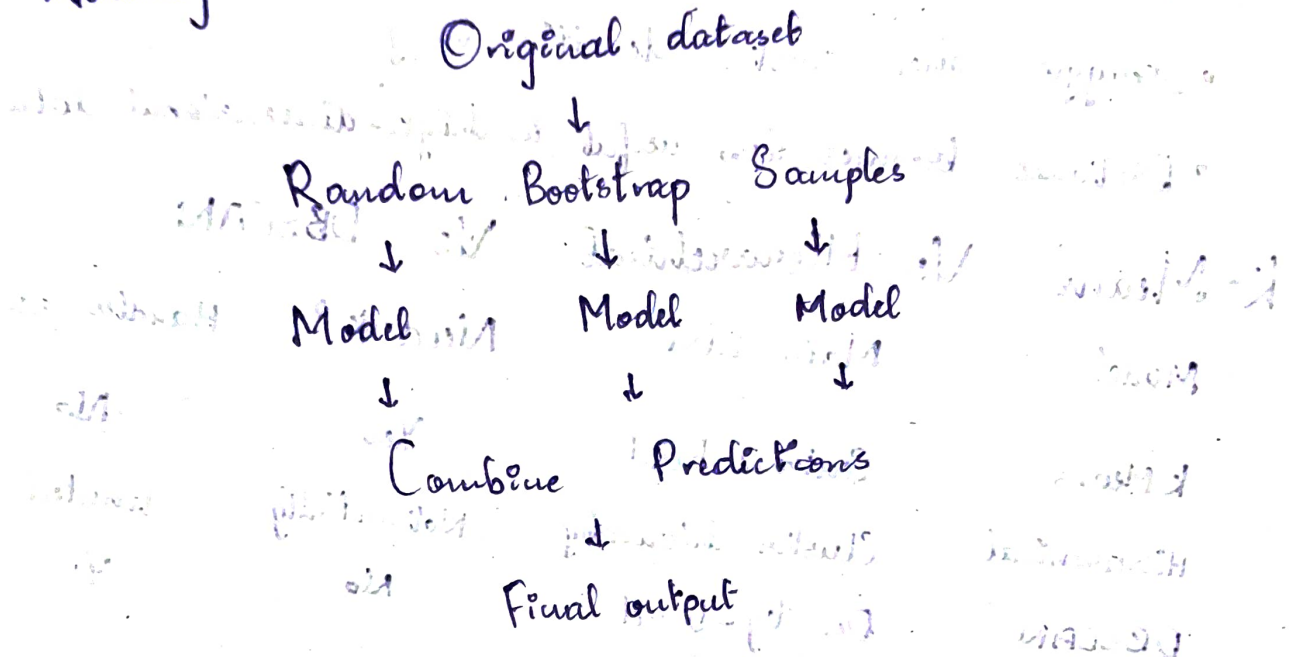
Main ensemble approaches:

- Bagging
- Boosting

8. Bagging - An ensemble method that trains multiple models independently on diff random samples of the training data and combines their predictions.

Bagging $\Rightarrow$ Bootstrap Aggregating

Working -



For classification - Majority voting

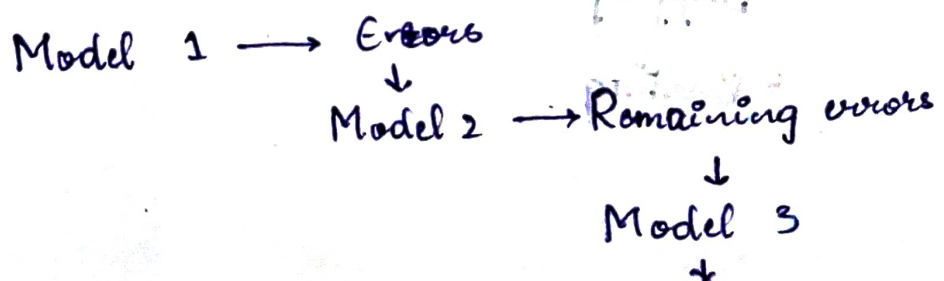
For regression - Average Prediction

Main purpose :

- Reduce variance
- Reduce overfitting

* Random Forest is a classic bagging-based algorithm.

9. Boosting - An ensemble method that trains models sequentially, where each new model focuses more on correcting errors made by previous models.



Combined prediction

Main purpose :

- Reduce bias
- Build a stronger learner

Common boosting algorithms :

- AdaBoost
- Gradient Boosting
- XGBoost

Bagging	Boosting
◦ Models trained mostly independently ◦ Reduce variance ◦ Less sensitive to individual errors ◦ Random Forest	◦ Models trained sequentially ◦ Mainly reduces bias ◦ Later models focus on earlier errors ◦ AdaBoost / Gradient Boosting / XGBoost

10. AdaBoost - A boosting algorithm that sequentially trains weak learners and gives more importance to incorrectly classified observations

AdaBoost stands for Adaptive Boosting

Working -

Train weak model



find misclassified points



increase their importance



train next model



Combine Models

Common weak learner:

Decision Stump - is a decision tree with one split.

Advantages -

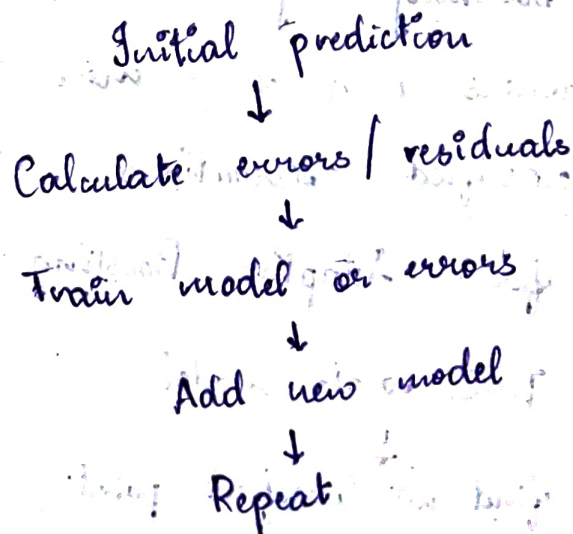
- Simple boosting technique
- Can improve weak learners significantly
- Works well on clean datasets

Limitations -

- Sensitive to noisy data
- " outliers
- Sequential training can be slower than bagging.

11. Gradient Boosting - A boosting method where each new model is trained to correct the residual errors of the previous ensemble.

Working -



Conceptually:

Prediction + correction + com + correction

Important parameters -

- No of estimators
- Learning rate
- Tree depth
- Subsampling

Advantages -

- Strong predictive performance
- Handles nonlinear relationships
- Works with many structured data problems

Limitations -

- Can overfit
- Sensitive to hyperparameters
- Training is sequential

12. XGBoost - An optimized implementation of gradient boosted decision trees designed for high performance, regularization and computational efficiency.

XGBoost $\Rightarrow$ Extreme gradient boosting

Main improvements -

- Regularization
- Efficient tree construction
- Parallelized parts of tree building
- Pruning
- Handling missing values
- Learning-rate control

Advantages -

- Excellent performance on tabular data
- Handles nonlinear relationships
- Supports regularization

• Often strong in competitions and practical ML

Limitations -

- More hyperparameters.
- Can be computationally expensive
- Less interpretable than simple models.

1. AdaBoost → focus more on misclassified observations.

2. Gradient Boosting → fit new models to residual errors

3. XGBoost → Optimized and regularized gradient boosting.

13. Semi-Supervised learning - A machine learning approach that trains using a small amount of labeled data together with a larger amount of unlabeled data.

small labeled dataset + large unlabeled dataset
└──────────────────┘
Semi-supervised learning

↓

Improved model

Useful when labeling data is ↓

- Expensive
- Slow
- Difficult
- Requires experts

Examples:

Medical images

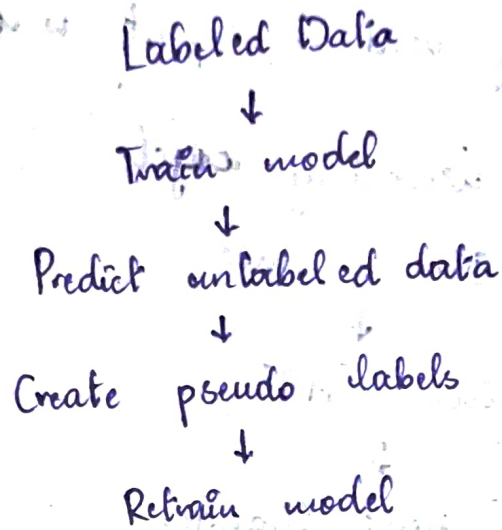
Speech recognition

Web page classification

Image

"

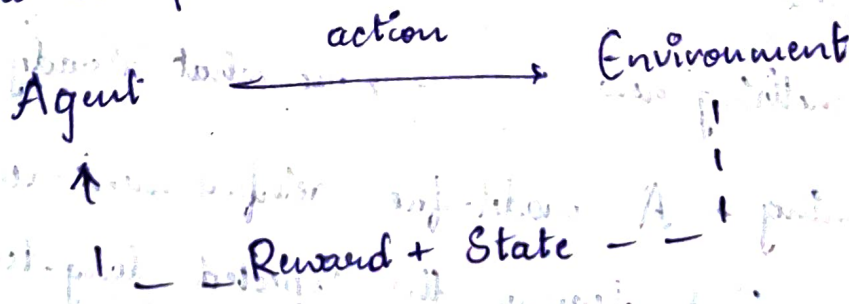
14. Pseudo-labeling - A semi-supervised technique where a model predicts labels for unlabeled examples and high-confidence predictions are reused as training labels.



15. Self-training - A semi-supervised method where a model repeatedly labels unlabeled examples and retrains itself using confident predictions.

→ pseudo-labeling is common form of self-training.

16. Reinforcement learning - A machine-learning approach where an agent learns which actions to take by interacting with an environment and receiving rewards or penalties.



Components -

- Agent: The learner or decision-maker.
- Environment: The world in which the agent operates.

- State - Current situation of the env.
- Action - A decision taken by the agent.
- Reward - Feedback indicating how good/bad an action was.
- Policy - Strategy used by agent to choose actions.

→ Reinforcement learning cycle

Observe state



Choose Action



Environment Changes



Receive reward



Learn



Repeat

17. Exploration

Exploitation

→ Trying new actions to discover potentially better strategies.

→ Choosing actions already known to produce high rewards.

→ learn something new

→ use what already works

18. Q-learning - A model-free reinforcement-learning algorithm that learns the expected long-term value of taking an action in a state.

→ $Q(s, a)$ = how useful action (a) is in state (s).

19. Reinforcement - learning examples:
- Game playing
 - Robotics
 - Autonomous Systems
 - Recommendation Systems
 - Resource Allocation